Applications of CA. Algebraic Number theory

I. Rings of Integers 1. Dedekind's theorem

Thm I.1.1. Let L be a field containing A. $\alpha \in L$ is integral over A iff $\exists$ nonzero finitely-generated A-submodule of L such that $\alpha M \subseteq M$.

proof: We've proved this in the course of integral dependence using Cayley-Hamilton theorem $\square$ (Coro VII.1.5).

Def I.1.2 The integral closure of $\mathbb{Z}$ in the algebraic number field L is called the ring of integers in L, denoted by $\mathcal{O}_L$.

Thm I.1.3. A unique factorization domain A is integral closed, namely, is its own closure in its fractorization field. (Gauss lemma).

proof: Let $\frac{a}{b} \in \mathrm{Frac}(A)$ integral over A. that is, $(\frac{a}{b})^n + a_1(\frac{a}{b})^{n-1} + \dots + a_n = 0$. $a_i \in A$.
$\Rightarrow a^n + a_1 a^{n-1} b + \dots + a_n b^n = 0$. (*) Suppose that $\frac{a}{b} \notin A$, then $\exists p$ s.t. $p \mid b$, $p \nmid a$
then apply this to (*), contradiction $\square$.

Examples: (1) $\mathbb{Z}$, $\mathbb{Z}[i]$ are integrally closed

(2) $\mathbb{Z}[\sqrt{-3}]$ isn't a UFD, and also we see $(\frac{-1+\sqrt{-3}}{2})^3 - 1 = 0$ which is integral over $\mathbb{Z}[\sqrt{-3}]$ does not belong to $\mathbb{Z}[\sqrt{-3}]$.
Moreover, $\mathrm{cl}(\mathbb{Z}[\sqrt{-3}]) = \mathbb{Z}[\omega]$ where closure is in $\mathbb{Q}[\sqrt{-3}]$, $\omega = \frac{-1+\sqrt{-3}}{2}$.

Thm I.1.4. $K = \mathrm{Frac}(A)$, and L/K of finite degree, A is integrally closed. α of L is integral over A iff ~~its~~ $m_{\alpha,K}(x)$ has coefficients in A.

proof: We've proved this in Prop VII.3.4. $\square$

Examples: $\alpha \in \mathbb{Z}[\sqrt{d}]$ is integral over $\mathbb{Z}$ iff its trace and norm lie in $\mathbb{Z}$.

Prop. I.1.5. B integral over A and finitely generated as an A-algebra, then it is finitely generated as an A-module. (finitely generated A-algebra $B := \{y \in B \mid y \in A[\beta_1, \dots, \beta_m], \beta_i \in B\}$)

proof: (skipped) (Start from $B = A[\beta]$).

Prop. I.1.6. $A \subseteq B \subseteq C$ integral domains, B integral over A, C integral over B. then C integral over A.

proof: Use the last Prop $\square$.

Coro I.1.7. $K = \mathrm{Frac}(A)$, L/K algebraic extension. $\mathrm{cl}(A)$ in L is integrally closed.

Examples: $\mathcal{O}_L$ is integrally closed

2. Discriminants.

Def I.2.1 {Trace / Norm} $A \subseteq B$ be rings s.t. B is a free A-module of rank n. Then $\forall \beta \in B$ defines a A-linear map $x \mapsto \beta x \quad B \to B$

If $\{e_i\}$ be the basis, $\beta e_i = \sum_{i,j=1}^{n} a_{ij} e_j$

Then $Tr_{B/A}(\beta) := \sum_{i=1}^{n} a_{ii}$.

$Nm_{B/A}(\beta) := \det(a_{ij})$.

Prop. I.2.2. $Tr(\beta+\beta') = Tr(\beta) + Tr(\beta')$; $Nm(\beta\beta') = Nm(\beta)\,Nm(\beta')$

$\alpha \in A$ $\quad Tr(\alpha\beta) = \alpha\, Tr(\beta) \qquad Nm(\alpha) = \alpha^n$

$Tr(\alpha) = n\alpha$

Def I.2.3 (1) If L is a finite extension of K (field). then $L \times L \to K$. $(\alpha, \beta) \mapsto Tr_{L/K}(\alpha\beta)$ is a symmetric bilinear form on L. , the discriminant of L/K is : $\det(Tr_{L/K})$

(2) L/K finite & seperable $\alpha_1, \dots, \alpha_n \in L$. Define $D(\alpha_1, \dots, \alpha_n) = \det\left(Tr_{L/K}^{(\alpha_i\alpha_j)}\right)$.

Prop. I.2.4. Let L/K finite & seperable, $\sigma_1, \dots, \sigma_n$ be the K-embedding : $L \to \mathbb{C}$.

Then $D(\alpha_1, \dots, \alpha_n) = \det(\sigma_i(\alpha_j))^2$.

pf: Since $Tr_{L/K}(\alpha_i\alpha_j) = \sum_{k=1}^{n} \sigma_k(a_i a_j)$ □.

Prop. I.2.5. $D(\alpha_1, \dots, \alpha_n) \neq 0 \iff \alpha_1, \dots, \alpha_n$ are linearly-independent.

pf: (skipped)

Prop. I.2.6. Let $L = K(\alpha)$, $[L:K] = n$. $f(x)$ be the minimal poly of α over K

$\alpha_1 = \alpha$, $\alpha_2, \dots \alpha_n$ be the roots of $f(x)$

Then $D(1, \alpha, \dots, \alpha^{n-1}) = \prod_{1 \le r < s \le n} (\alpha_r - \alpha_s)^2 = (-1)^{\frac{n(n-1)}{2}} N_{L/K}(f'(\alpha))$.

Rmk. Here we skip the evquivalence between: $Tr_{L/K}(\beta) = \sum_{i=1}^{n} a_{ii}$. (The general definition

$\sigma_i : L \to \mathbb{C}$ be K-embedding. $Tr_{L/K}(\beta) = \sum_{i=1}^{n} \sigma_i(\beta)$ (The def in field embedding